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SELF-CONTAINED BEARING AND WIRELESS POWER TRANSMISSION DEVICE

Abstract

A bearing assembly includes an annular carrier, an outer ring disposed within the carrier, the outer ring having an outer raceway and a transmitter coil spaced axially from the outer raceway. The transmitter coil is electrically connectable with a battery and is configured to generate an electromagnetic field when current flows through the transmitter coil. An inner ring is disposeable about a rotor shaft and has an inner raceway and a receiver coil spaced axially from the inner raceway so as to be located proximal to and inductively coupled with the transmitter coil such that electric current is generated within the receiver coil when current flows through the transmitter coil. The receiver coil is connected to electromagnetic coils of a motor rotor such that electric current flows from receiver coil to the coils. A plurality of rolling elements are disposed between rotatably couple the outer ring with the inner ring.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates to both bearings and power transmission devices, and more particularly to power transmission devices for electric motors with shafts supported by bearings.

[0002] Electric motors are well known and include a stator with one or more magnets, typically electromagnets, and a rotor disposed within the stator, mounted on or connected to a shaft and including one or more magnets. When the rotor magnets are electromagnetic coils, electric current or power must be supplied to these coils in order to generate the required magnetic fields which interact with corresponding magnetic fields of the stator. Typically, electric power is supplied to the rotor electromagnet coils by means of one or more fixed conductors electrically connected with a power supply, for example a battery, and rotatable conductors mounted on the shaft and electrically connected with the rotor electromagnet(s) and contacted by the fixed conductor(s).

[0003] As the rotor shaft angularly displaces about a central axis, the rotatable conductors slide against the fixed conductors to continuously transfer current from the power supply to the rotor electromagnetic coil(s). Due to such sliding motion, both the fixed and rotatable conductors may experience wear, such that the power transfer between the conductors is reduced or even entirely disrupted and correspondingly reduce or cease the current flow to the rotor electromagnetic coils. Also, debris or contaminants may become disposed or “lodged” in the interface between the conductors with a resultant diminishment or cessation of current flow to the rotor. Further, the sliding contact between the conductors increases the friction loading on the rotor shaft.

[0004] Due to these potential issues, a typical motor used in an electric vehicle has a rotor with permanent magnets so that there is no need to supply current to the rotor. Although such a power system is relatively robust, the permanent magnets required to generate sufficient torque to drive an electric vehicle are substantially large and thereby significantly increases the weight of the vehicle. Further, such large permanent magnets are typically formed of rare earth elements such as neodymium, terbium, dysprosium, etc., such that detrimental environmental impacts may result from the mining, processing and disposal of such materials.

SUMMARY OF THE INVENTION

[0005] In one aspect, the present invention is a bearing assembly with a power transmission device for transmitting electric power from a battery to an electric motor. The motor has a housing, an outer stator connected with the housing and an inner rotor mounted to a shaft, disposed within the stator and rotatable about a central axis extending through the shaft, the rotor having at least one electromagnetic coil. The bearing assembly comprises an annular carrier connectable with the housing and having a bore and an outer ring disposed within the bore of the carrier, the outer ring having an outer raceway and a transmitter coil spaced axially from the outer raceway. The transmitter coil is electrically connectable with the battery and configured to generate an electromagnetic field when current flows through the transmitter coil. An inner ring is disposeable about the shaft and has an inner raceway and a receiver coil spaced axially from the inner raceway so as to be located proximal to and inductively coupled with the transmitter coil. As such, electric current is generated within the receiver coil when current flows through the transmitter coil.

Further, the receiver coil is electrically connectable with the at least one electromagnetic coil of the rotor, such that electric current flows from receiver coil to the electromagnetic coil. Furthermore, a plurality of rolling elements are disposed between the outer raceway and the inner raceway and rotatably coupling the outer ring with the inner ring to thereby couple the rotor shaft with the housing.

[0006] In another aspect, the present invention is also a drive motor assembly for an electric vehicle having at least one wheel. The motor assembly comprises an electric motor configured to

rotatably drive the at least one wheel, the motor having a housing, an outer stator connected with the housing and an inner rotor. The rotor is mounted on a shaft, disposed within the stator and rotatable about a central axis extending through the shaft, the rotor having at least one electromagnetic coil. The motor assembly further comprises a battery and a bearing assembly with a power transmission device as described in the preceding paragraph.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0007] The foregoing summary, as well as the detailed description of the preferred embodiments of the present invention, will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there is shown in the drawings, which are diagrammatic, embodiments that are presently preferred. It should be understood, however, that the present invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:

[0008] FIG. 1 is a more diagrammatic view of an electric vehicle having a bearing assembly with a power transmission device in accordance with the present invention;

[0009] FIG. 2 is a block diagram of certain electronic components of the drive system of the vehicle, showing the flow of electric power through the vehicle;

[0010] FIG. 3 is an axial cross-sectional view of the drive system including the bearing assembly of the present invention, shown installed about a shaft and connected with a housing;

[0011] FIG. 4 is a broken-away, enlarged view of a portion of FIG. 3; and

[0012] FIG. 5 is broken-away, more enlarged view of a portion of FIG. 3.

DETAILED DESCRIPTION OF THE INVENTION

[0013] Certain terminology is used in the following description for convenience only and is not limiting. The words “inner”, “inwardly” and “outer”, “outwardly” refer to directions toward and away from, respectively, a designated centerline or a geometric center of an element being described, the particular meaning being readily apparent from the context of the description. Further, as used herein, the words “connected” and “coupled” are each intended to include direct connections between two members without any other members interposed therebetween and indirect connections between members in which one or more other members are interposed therebetween. The terminology includes the words specifically mentioned above, derivatives thereof, and words of similar import.

[0014] Referring now to the drawings in detail, wherein like numbers are used to indicate like elements throughout, there is shown in FIGS. 1-5 a self-contained bearing assembly **10** with an incorporated wireless power transmission device **11** for transmitting electric power from a battery **1**, which is preferably a battery pack, to an electric motor **2**, preferably through an electronic power inverter attached to the motor **2**. The motor **2** is preferably a drive motor **2** of an electric vehicle **V** with four wheels **W**, as depicted in FIG. 1, but may alternatively be any other type of electric motor. The motor **2** has a housing **3**, an outer stator **4** connected with the housing **3** and an inner rotor **5**. The rotor **5** is mounted to a shaft **6**, disposed within the stator **4** and rotatable about a central axis **Ac** extending through the shaft **6**. The rotor **5** has at least one and preferably a plurality of electromagnetic coils **7**. Basically, the bearing assembly **10** with the power transmission device **11** comprises a carrier **24**, an outer ring **12** with a transmitter coil **14** disposed within the carrier **24**, an inner ring **16** with a receiver coil **18** disposed within the outer ring **12** and a plurality of rolling elements **20** disposed between and rotatably coupling the inner and outer rings **12**, **16**. Electric power is wirelessly transferred from the transmitter coil **14** to the receiver coil **18** during rotation of the inner ring **16**, as described in detail below. Further, by providing the carrier **24**, the bearing assembly **10** is “self-contained” as described in detail below.

[0015] Referring to FIGS. 3-5, the carrier **24** is annular and is connectable with the housing **3**, preferably disposeable within a bore **8** of the housing **3**. The carrier **24** has a bore **25** and the bearing outer ring **12** is disposed within the bore **25** of the carrier **24**. Preferably, the carrier **24** includes a tubular portion **26**, which is disposable within the housing bore **8** and has an inner circumferential surface **26a** defining the carrier bore **25**, a mounting flange portion **27A** extending radially outwardly from the tubular portion **26** and a retainer flange portion **27B** extending radially inwardly from the tubular portion **26**. The mounting flange portion **27A** is disposable against a radial surface **3a** of the housing **3** and is connectable with the housing **3**, preferably by means of one or more fasteners **64**. However, the carrier **24** may alternatively be connected to the housing **3** by any other appropriate means, such as for example, by means of a threaded connection.

[0016] Preferably, the outer ring **12** is slidably disposed within the carrier bore **25** so as to enable axial displacement of the outer ring **12** to accommodate or adjust to thermal expansion of the shaft **6**. More specifically, the carrier **24** includes an axial retainer **28**, which is preferably formed as an annular shoulder of the tubular portion **26** and is disposeable against a second axial end **30b** (described below) of the outer ring **12**. At least one biasing member **29** is disposed within the carrier bore **25** and has a first axial end **29a** disposed against the retainer flange portion **27B** of the carrier **24** and a second axial end **29b** disposed against the first axial end **30a** (described below) of the outer ring body **30**. The biasing member(s) **29** are each configured to exert an axial force on the first axial end **30a** of the outer ring **12** in an axial direction toward the retainer **28**. As such, the outer ring **12** is capable of limited axial displacement but an appropriate preload is always established within the rolling elements **20**.

[0017] Further, the at least one biasing member **29** is preferably a single wave spring (not depicted), but may alternatively include a plurality of coil springs (as shown), a solid elastomeric member(s), etc. However, the bearing outer ring **12** may alternatively be fixedly disposed within the carrier bore **25**, such as by means of a friction fit between the bore inner surface **26a** and a ring outer surface **34** (described below), in which case the bearing assembly **10** would, of course, not include any biasing member.

[0018] Still referring to FIGS. 3-5, the outer ring **12** has an outer raceway **13**, the transmitter coil **14** being attached to the outer ring **12** and spaced axially from the outer raceway **13**. The transmitter coil **14** is electrically connectable with the battery **1**, preferably through one or more electronic components as discussed below, and is preferably configured to generate an electromagnetic field. The inner ring **16** is disposeable or disposed about the shaft **6** and has an inner raceway **17**, the receiver coil **18** being attached to the inner ring **16** and spaced axially from the inner raceway **17** so as to be located proximal to and magnetically coupled with the transmitter coil **14**. The receiver coil **18** is electrically connectable with the at least one electromagnetic coil **7** of the rotor **5** such that current generated within the receiver coil **18** flows to the electromagnetic coil(s) **18**, preferably through one or more electronic components described below.

[0019] More specifically, the transmitter coil **14** receives electric current and generates an electromagnetic field which induces an electric current within the receiving coil **18**, which is referred to as "inductive coupling" and is well known in the field of electronics, such that a detailed explanation of the coils **14**, **18** is beyond the scope of the present disclosure. In order to provide sufficient electric current to the rotor **5** in the preferred application of an electric vehicle **V**, the transmitter coil **14** is configured, i.e., constructed, sized, etc., to have the capacity to transmit an amount of electrical power within a range of one hundred watts (100 W) and three kilowatts (3 kW) and the receiver coil **18** is similarly configured to receive an amount of electrical power within the range of one hundred watts and (100 W) three kilowatts (3 kW). However, the two coils **14**, **18** may be sized or formed with a substantially lower or substantially greater capacity to transmit electric power depending on the specific application of the bearing assembly **10**.

[0020] Further, the plurality of rolling elements **20** are disposed between the outer raceway **13** and the inner raceway **17** and rotatably couple the outer ring **12** with the inner ring **16** and forming a

bearing **22**. As depicted, the rolling elements **20** are preferably balls, but may alternatively be cylindrical rollers, tapered rollers, needles or any other appropriate type of rolling element. The rolling elements **20** may be conventionally formed of steel, but may alternatively be formed of ceramic or another insulative material so as to provide insulation between the bearing raceways **13**, **17**. Although not presently preferred, the bearing rings **12**, **16** may even alternatively be formed as components of a plain bearing with an inner circumferential surface of the outer ring **12** being slidably disposed about an outer circumferential surface of the inner ring **16**, the transmitter and receiver coils **14**, **18** being appropriately positioned within the rings **12**, **16** (structure not shown). [0021] Preferably, the outer ring **12** includes an annular body **30** with opposing first and second axial ends **30a**, **30b**, respectively, an inner circumferential surface **32** and an opposing outer circumferential surface **34**. The outer raceway **13** extends radially outwardly from the inner circumferential surface **32** and the transmitter coil **14** is either disposed directly on the inner circumferential surface **32** or, preferably, within an annular recess **36** extending radially outwardly from the inner circumferential surface **32**. Further, the outer circumferential surface **34** is slidably disposed within a bore **8** of the housing **3** and has an annular groove **35** extending radially inwardly from the outer surface **34**, for reasons discussed below.

[0022] Similarly, the inner ring **16** includes an annular body **40** with opposing first and second axial ends **40a**, **40b**, respectively, an inner circumferential surface **42** and an opposing outer circumferential surface **44**. The inner raceway **17** extends radially inwardly from the outer circumferential surface **44** and the receiver coil **18** is either disposed directly on the outer circumferential surface **44** or, preferably, within an annular recess **45** extending radially inwardly from the outer circumferential surface **44**. Further, the inner circumferential surface **42** defines an annular bore **46** which receives a portion of the shaft **6**, the inner ring **16** preferably being frictionally engaged with the shaft **6**.

[0023] Preferably, the annular body **40** of the inner ring **16** has an axial length (not indicated) between the ends **40a**, **40b** that is substantially greater than an axial length (not indicated) between the ends **30a**, **30b** of the body **30** of the outer ring **12**. As such, an exterior mounting surface section **48** of the outer circumferential surface **42** extends between the second axial end **40b** and the receiver coil **18**. As described below, the mounting surface section **48** provides mounting surfaces for certain electronic components connected with the receiver coil **18**.

[0024] In a presently preferred embodiment, each one of the transmitter coil **14** and the receiver coil **18** is annular, with the receiver coil **18** being disposed coaxially within the transmitter coil **14**. More specifically, the transmitter coil **14** has an annular body **15** with an inner circumferential surface **15a** and the receiver coil **18** has an annular body **19** with an outer circumferential surface **19a** spaced radially inwardly from the inner circumferential surface **15a** of the transmitter coil **14**. As such, an annular air gap GA (FIG. 5) is defined between the two surfaces **15a**, **19a**, as indicated in FIG. 5. Preferably, transmitter coil **14** and the receiver coil **18** are each sized such that a radial dimension $d_{sub.R}$ of the annular air gap GA has a value of no greater than five millimeters (5 mm) and preferably about one millimeter (1 mm). With such a relatively small or narrow air gap GA, the power transmission device **11** is capable of transferring a relatively substantial amount of electric power, as discussed above, with a relatively high efficiency, preferably with an efficiency of greater than ninety-five percent (>95%).

[0025] Alternatively, the transmitter coil **14** and the receiver coil **18** may again be formed annular, but spaced axially apart and having facing axial end surfaces (not shown). As another alternative, each one of the transmitter coil **14** and the receiver coil **18** may be formed with facing angled surfaces that are spaced both radially and axially apart. Further, both the transmitter coil **14** and the receiver coil **18**, as well as supporting electronic components for the coils **14**, **18**, may be formed or configured to provide bidirectional signal or data transmission. For example, data may be transferred from the rotor **5** (e.g., taken by sensors) to a non-rotating component (e.g., a motor controller, vehicle computer, etc.) and provide information on the rotor **5**, such as temperature, etc.,

to a vehicle control system (not shown). As another example, a vehicle control system can provide instructions to adjust a rectifier system (not shown) by sending data or commands from the transmitter coil **14** to the receiver coil **18**, the receiver coil **18** being connected with the rectifying system. The scope of the present invention encompasses all of these disclosed structures of the coils **14**, **18** and any other appropriate structure that is disposeable within or on the bearing rings **12**, **16** and capable of functioning as described herein.

[0026] Referring now to FIGS. **2**, **4** and **5**, the battery **1** provides direct current (DC) and the preferred electromagnetic coil(s) **7** of the rotor **5** require direct current. However, the transmitter coil **14** requires alternating current (AC) and generates alternating current within the receiver coil **18**. As such, the power transmission device **11** further comprises a direct current to alternating current (DC-to-AC) power converter **50**, which is preferably mounted to the housing **3**, and an alternating current to direct current (AC-to-DC) power converter **52**, which is preferably mounted to the inner ring **16**. More specifically, the DC-to-AC power converter **50** is electrically connectable/connected with the battery **1** and is electrically connected with the transmitter coil **14**, for example connected to each by means of a wire **54**, which may be hard-wired or removably connected (e.g., a plug-socket connection). The DC-to-AC power converter **50** is configured to convert direct current from the battery **1** to alternating current and to transmit the alternating current to the transmitter coil **14**. Preferably, the DC-to-AC converter **50** is disposed on a printed circuit board (PCB) **56** connectable or connected to the housing **3**. Most preferably, the PCB **56** is disposed on the mounting flange portion **27** of the carrier **24**, preferably by the same fastener(s) connecting the carrier **24** with the housing **3**.

[0027] Further, the AC-to-DC power converter **52** is preferably disposed on the mounting surface section **48** of the inner ring **16** and is electrically connected with the receiver coil **18** and electrically connectable or connected with the at least one electromagnetic coil **7** of the rotor **5**. The AC-to-DC power converter **52** is configured to convert alternating current generated within the receiver coil **16** to direct current transmittable to the electromagnetic coil(s) **7**. Preferably, the AC-to-DC converter **52** is disposed on a printed circuit board (PCB) **58A** mounted on the inner ring **16**, most preferably on the mounting surface section **48**. The PCB **58A** is preferably formed as a "horseshoe" PCB with a rectifying system and is retained on the mounting surface **48** by means of an annular retainer **59A** engaged with the surface section **48**, for example by friction, threading or a key. Further, the AC-to-DC power converter **52** is electrically connected with the electromagnetic coils **7** by any appropriate means, such as for example by two wires **54** (only one shown) connected with the converter **52**, the wires **54** being hard-wired or removably connected (e.g., by detachable contacts). Furthermore, the bearing assembly **10** may further comprise a second printed circuit board **58B** disposed on the mounting surface **48** of the inner ring **16**, and retained by a second retainer **59B**, for providing additional functionality to the bearing assembly **10**.

[0028] The bearing assembly **10** of the present invention has a number of advantages over previously known bearings for electric motors and power transmission devices for transmitting current to the motor rotor. By incorporating the power transmission device **11** within the bearing assembly **10**, the power transmission components, e.g., the transmission and receiver coils **14**, **18**, the power converters **50**, **52**, etc., are installed simultaneously with the installation of the bearing **22** supporting the rotor shaft **6**. Specifically by providing the bearing **22** and the components of the power transfer device **11** disposed within or attached to the carrier **24**, the bearing **22** and transfer device **11** are installed by simply inserting the rotor shaft **6** through the bearing inner ring **16** and attaching the carrier **24** to the housing radial surface **3a**.

[0029] As such, a separate installation process for the power components is eliminated and the axial space for the power components is reduced. Further, by providing electric power from the battery **1** to the rotor **5** by means of a wireless power transfer device **11**, the friction loading on the shaft **6** is reduced and both reliability and power density of the power transfer is substantially increased in comparison to transfer by contacting conductors. Also, the transmitter coil **14** and the receiver coil

18 are enclosed within the bearing rings **12, 16**, thereby providing substantial protection to these components **14, 18** and eliminating the need for a separate housing.

[0030] Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention.

[0031] Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

[0032] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter. The invention is not restricted to the above-described embodiments, and may be varied within the scope of the following claims.

Claims

1. A bearing assembly with a power transmission device for transmitting electric power from a battery to an electric motor, the motor having a housing, an outer stator connected with the housing and an inner rotor mounted to a shaft, disposed within the stator and rotatable about a central axis extending through the shaft, the rotor having at least one electromagnetic coil, the bearing assembly comprising: an annular carrier connectable with the housing and having a bore; an outer ring disposed within the bore of the carrier and having an outer raceway and a transmitter coil spaced axially from the outer raceway, the transmitter coil being electrically connectable with the battery and configured to generate an electromagnetic field when current flows through the transmitter coil; an inner ring disposeable about the shaft and having an inner raceway and a receiver coil spaced axially from the inner raceway so as to be located proximal to and inductively coupled with the transmitter coil such that electric current is generated within the receiver coil when current flows through the transmitter coil, the receiver coil being electrically connectable with the at least one electromagnetic coil of the rotor such that electric current flows from receiver coil to the electromagnetic coil; and a plurality of rolling elements disposed between the outer raceway and the inner raceway and rotatably coupling the outer ring with the inner ring to thereby couple the rotor shaft with the housing.
2. The bearing assembly as recited in claim 1, wherein the transmitter coil is annular and has an inner circumferential surface, the receiver coil is annular, is disposed within the transmitter coil and has an outer circumferential surface, an annular air gap being defined between the inner circumferential surface of the transmitter coil and the outer circumferential surface of the receiver coil.
3. The bearing assembly as recited in claim 2, wherein the transmitter coil and the receiver coil are each sized such that a radial dimension of the annular air gap has a value of no greater than five millimeters (5 mm).
4. The bearing assembly as recited in claim 1, wherein the transmitter coil is configured to transmit an amount of electrical power within a range of one hundred watts and three kilowatts and the receiver coil is configured to receive an amount of electrical power within the range of one hundred

watts and three kilowatts.

5. The bearing assembly as recited in claim 1, wherein: the outer ring has an inner circumferential surface, the outer raceway extending radially outwardly from the inner circumferential surface and the transmitter coil is disposed on the inner circumferential surface or on an annular recess extending radially outwardly from the inner circumferential surface; and the inner ring has an outer circumferential surface, the inner raceway extending radially inwardly from the outer circumferential surface and the receiver coil being disposed on the outer circumferential surface or an annular recess extending radially inwardly from the outer circumferential surface.

6. The bearing assembly as recited in claim 1, further comprising a DC-to-AC power converter mounted to the carrier, electrically connectable with the battery and with the transmitter coil and configured to convert direct current from the battery to alternating current and to transmit the alternating current to the transmitter coil.

7. The bearing assembly as recited in claim 6, wherein the DC-to-AC power converter is disposed on a printed circuit board connected with the carrier.

8. The bearing assembly as recited in claim 1 further comprising an AC-to-DC power converter mounted to the inner ring, the AC-to-DC converter is electrically connected with the receiver coil, electrically connectable with the at least one electromagnetic coil of the motor and configured to convert alternating current generated within the receiver coil to direct current transmittable to the electromagnetic coil.

9. The bearing assembly as recited in claim 8, wherein the wherein the AC-to-DC power converter is disposed on a printed circuit board mounted on the inner ring.

10. The bearing assembly as recited in claim 1, wherein the outer ring has first and second, opposing axial ends, the carrier includes an axial retainer disposeable against the second axial end of the outer ring and at least one biasing member disposed within the carrier bore and configured to exert an axial force on the first axial end of the outer ring in an axial direction toward the retainer.

11. The bearing assembly as recited in claim 10, wherein the carrier includes a tubular portion having an inner circumferential surface defining the carrier bore and a mounting flange portion extending radially outwardly from the tubular portion, the tubular portion being disposeable within a bore of the housing and the mounting flange portion being connectable to a radial surface of the housing.

12. A drive motor assembly for an electric vehicle having at least one wheel, the motor assembly comprising: an electric motor configured to rotatably drive the at least one wheel, the motor having a housing, an outer stator connected with the housing and an inner rotor, the rotor being mounted on a shaft, disposed within the stator and rotatable about a central axis extending through the shaft, the rotor having at least one electromagnetic coil; a battery; and a bearing assembly with a power transmission device for transmitting electric power from the battery to the motor and including: an annular carrier connectable with the housing and having a bore; an outer ring disposed within the bore of the carrier and having an outer raceway and a transmitter coil spaced axially from the outer raceway, the transmitter coil being electrically connectable with the battery and configured to generate an electromagnetic field when current flows through the transmitter coil; an inner ring disposed about the shaft and having an inner raceway and a receiver coil spaced axially from the inner raceway so as to be located proximal to and inductively coupled with the transmitter coil such that electric current is generated within the receiver coil when current flows through the transmitter coil, the receiver coil being electrically connected with the at least one electromagnetic coil of the rotor such that electric current flows from receiver coil to the electromagnetic coil; and a plurality of rolling elements disposed between the outer raceway and the inner raceway and rotatably coupling the outer ring with the inner ring to thereby rotatably couple the rotor shaft with the housing.

13. The motor assembly as recited in claim 12, wherein the transmitter coil is annular and has an inner circumferential surface, the receiver coil is annular, is disposed within the transmitter coil and

has an outer circumferential surface, an annular air gap being defined between the inner circumferential surface of the transmitter coil and the outer circumferential surface of the receiver coil.

14. The motor assembly as recited in claim 13, wherein the transmitter coil and the receiver coil are each sized such that a radial dimension of the annular air gap has a value of no greater than five millimeters (5 mm).

15. The motor assembly as recited in claim 12, wherein: the outer ring has an inner circumferential surface, the outer raceway extending radially outwardly from the inner circumferential surface and the transmitter coil is disposed on the inner circumferential surface or on an annular recess extending radially outwardly from the inner circumferential surface; and the inner ring has an outer circumferential surface, the inner raceway extending radially inwardly from the outer circumferential surface and the receiver coil being disposed on the outer circumferential surface or an annular recess extending radially inwardly from the outer circumferential surface.

16. The motor assembly as recited in claim 12, further comprising a DC-to-AC power converter mounted to the carrier, electrically connectable with the battery and with the transmitter coil and configured to convert direct current from the battery to alternating current and to transmit the alternating current to the transmitter coil.

17. The motor assembly as recited in claim 16, wherein the DC-to-AC power converter is disposed on a printed circuit board connected with the carrier.

18. The motor assembly as recited in claim 12, further comprising an AC-to-DC power converter mounted to the inner ring, the AC-to-DC converter is electrically connected with the receiver coil, electrically connectable with the at least one electromagnetic coil of the motor and configured to convert alternating current generated within the receiver coil to direct current transmittable to the electromagnetic coil.

19. The motor assembly as recited in claim 18, wherein the wherein the AC-to-DC power converter is disposed on a printed circuit board mounted on the inner ring.

20. The motor assembly as recited in claim 12, wherein the outer ring of the bearing is slidably disposed within the bore of the carrier.
